

Instructions:

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- The exam is to be solved on this paper.
- State your answers in lisible handwriting and with the help of diagrams, where necessary.
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Q1 While flying east with a speed of 200km/h , the main rotor blade of a helicopter is rotating at a speed of 500rpm . What is the linear velocity of a rotor blade tip with respect to ground and with respect to point A (figure 1), when it passes just in front of it. Length of the rotor blade (E) is 10m and height of rotor blade from ground (F) is 5m . Height of point A from ground is 1.5m . [10]

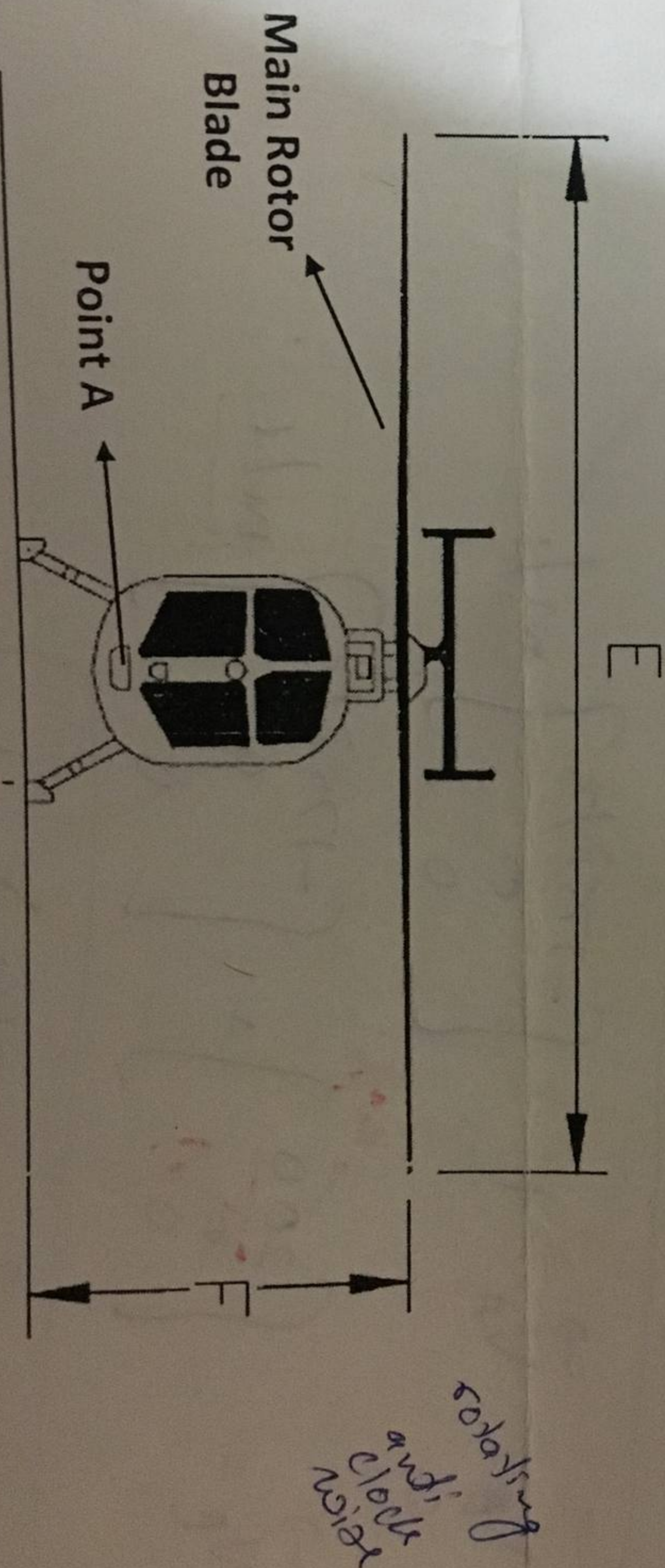


Figure 1 – Front View of a Helicopter

let rotor blade tip = C



Name: Fawad Arshad

Roll No.: 132-4411

8

Total Marks : 10

Time Allowed: 15 minutes

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Q1 The forces F_1 , F_2 , and F_3 , all of which act on point A of a bracket shown in figure 1. [10]
Determine the resultant force F_R . Resolve F_R along F_2 .

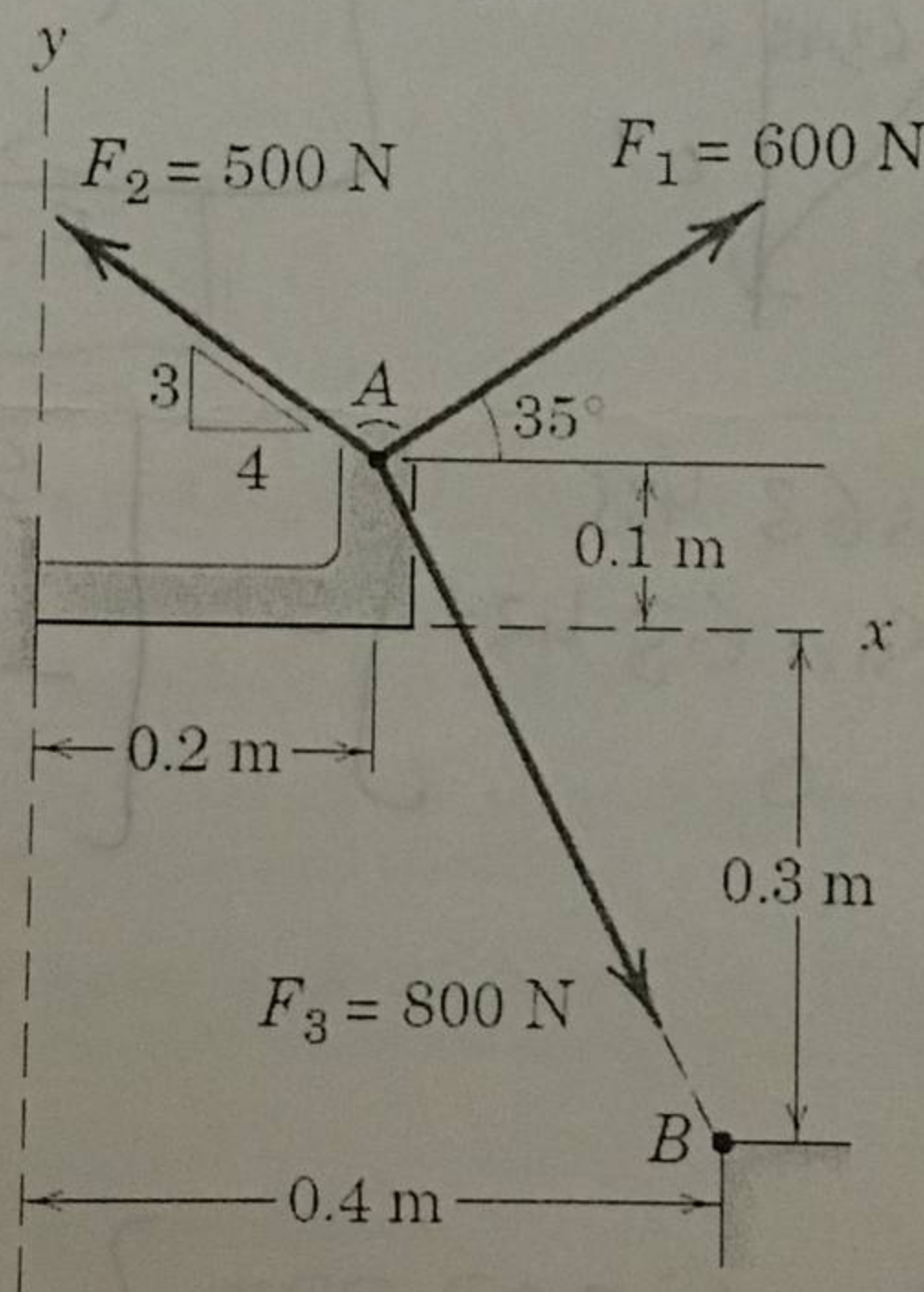


Figure 1 - A Loaded Bracket

Solution starts here.....

$$\vec{F}_R = \vec{F}_2 + \vec{F}_1 + \vec{F}_3 \quad \text{N}$$

resolving F_1 :

$$\vec{F}_1 = \begin{bmatrix} 600 \cos 35^\circ \\ 600 \sin 35^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} 491.491 \\ 344.146 \\ 0 \end{bmatrix} \text{ N}$$

$\vec{F}_2$:

$$\vec{F}_2 = \begin{bmatrix} -500 \cos 36.87^\circ \\ 500 \sin 36.87^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} -400 \\ 300 \\ 0 \end{bmatrix} \text{ N}$$

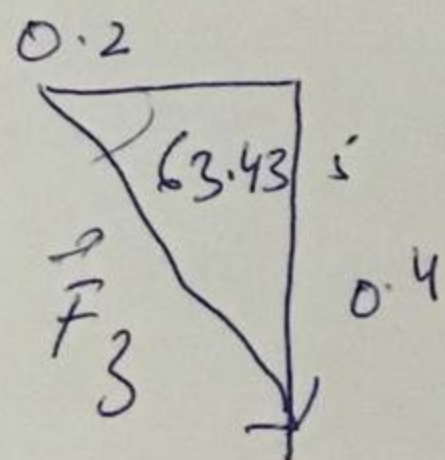
Projection of $\vec{F}_2$ along $\vec{F}_2$.

$$\hat{F}_2 =$$

$$\frac{\vec{F}_2 \cdot \vec{F}_2}{F_2}$$

$$\begin{bmatrix} 449.261 \\ -71.396 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -400 \\ 300 \\ 0 \end{bmatrix}^2$$

$\vec{F}_3$:



$$\vec{F}_{R \text{ along } F_2} = \begin{bmatrix} 357.770 \\ -25.702 \\ 0 \end{bmatrix} \text{ N}$$

$$\vec{F} - \vec{F}_{R \text{ along } F_2} = \vec{F}_{\perp \text{ to } F_2}$$

$$\vec{F}_3 = \begin{bmatrix} 800 \cos 63.435^\circ \\ -800 \sin 63.435^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} 357.770 \\ -715.542 \\ 0 \end{bmatrix} \text{ N}$$

$$F_R = 1$$

$$\vec{F}_R = \begin{bmatrix} -400 \\ 300 \\ 0 \end{bmatrix} + \begin{bmatrix} 357.770 \\ -715.542 \\ 0 \end{bmatrix} + \begin{bmatrix} 491.491 \\ 344.146 \\ 0 \end{bmatrix}$$

$$\vec{F}_R = \begin{bmatrix} 449.261 \\ -71.396 \\ 0 \end{bmatrix} \text{ N}$$

$$F_R = 454.898 \angle -9.029^\circ \text{ N}$$

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Q1 The blocks shown in Figure 1 are connected by flexible, inextensible cords passing over frictionless pulleys. At A the coefficients of friction is $\mu = 0.30$. Calculate the coefficient of friction at B, if the system is in static equilibrium. [10]

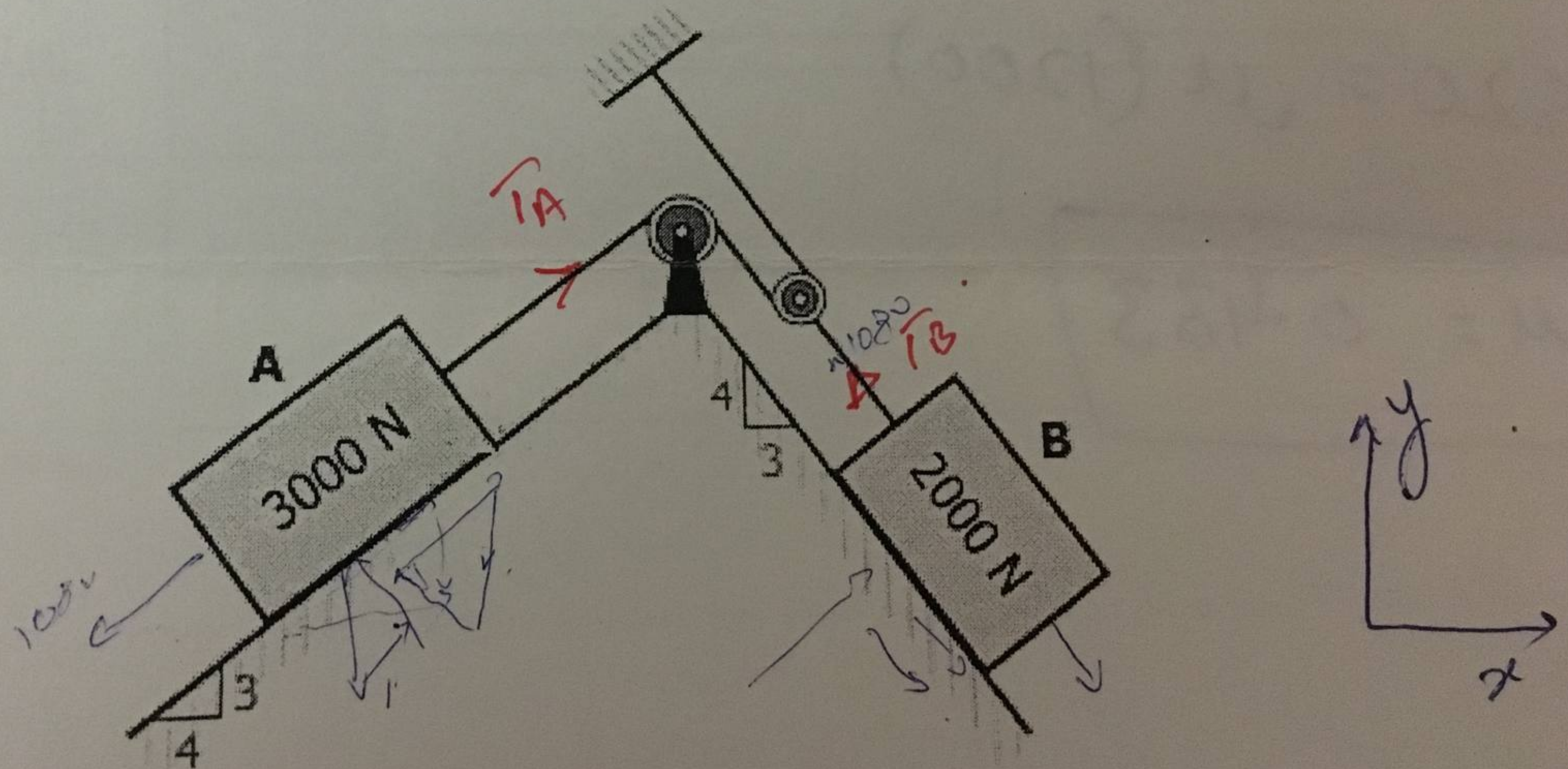


Figure 1 – Blocks on Inclined Plane

Name: *Fawad Arshad*

Roll No. : *13C-4411*

Total Marks : 10

Time Allowed: 20 minutes

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Q1 The pendulum shown in figure 1 consists of a plate having a weight of 12kg and a slender rod having a weight of 4kg . Determine moment of inertia of the pendulum about point O. Also find the center of gravity of the pendulum as a whole. [10]

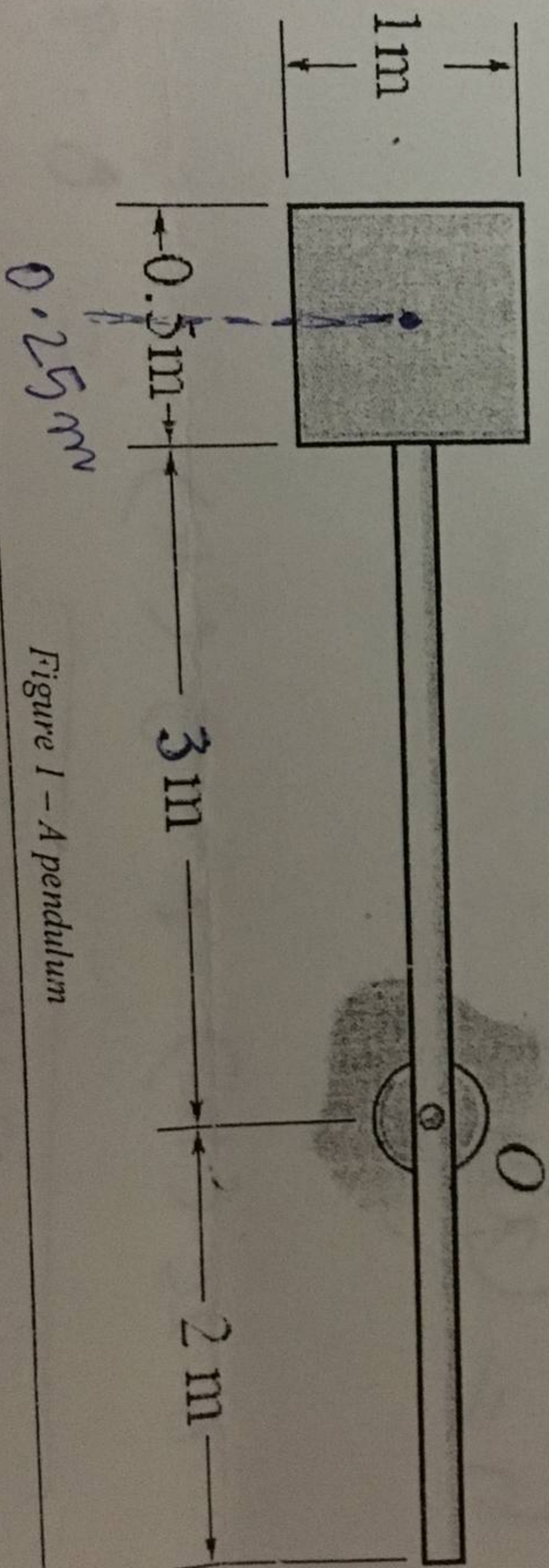


Figure 1 - A pendulum

Solution starts here.....



Q1 A 6kg slotted disk having three holes is shown in figure 1. Find the moment of inertia about its center of gravity along an axis perpendicular to the plane of the figure. [10]

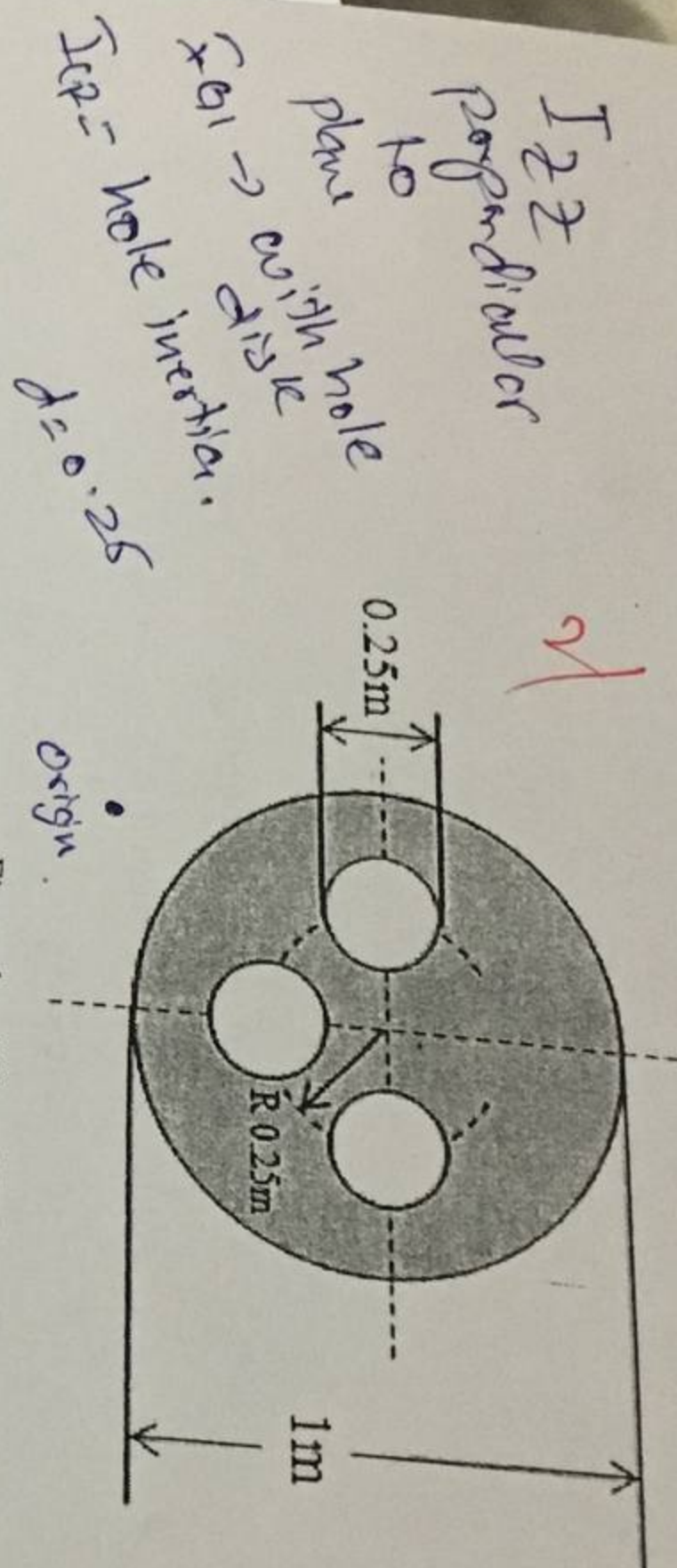


Figure 1 - A Slotted Disk

inertia:
m = 4.875
with hole

$$I_G = I_{G1} - I_{G2} - I_{G3}$$

$$I_{G1} = 0.75 + md^2$$

$$I_{G2} = 0.75 + 6 \times (0.014)^2$$

$$I_{G1} = 0.752$$

$$I_{G2} = 0.026$$

$$I_{G2} = I_{G3} = I_{G4}$$

Center of gravity
mass of holes:

mass of disk without holes = 6kg

mass density = $\frac{6}{\pi(0.5)^2} = 7.640 \text{ kg/m}^2$

Center of gravity:
 $x = 0.5(6) + 0.25(-0.375) + 0.5(-0.375) + 0.75(-0.375)$

$I_G = 0.674 \text{ kgm}^2$

$y = 0.5(6) + 0.5(-0.375) + 0.5(-0.375) + 0.25(0.375)$

$y = 0.519 \text{ m}$

$d = 0.019$

$6(0.5, 0.519)$



Name: Fawad Arshad

Roll No.: 13L-4411

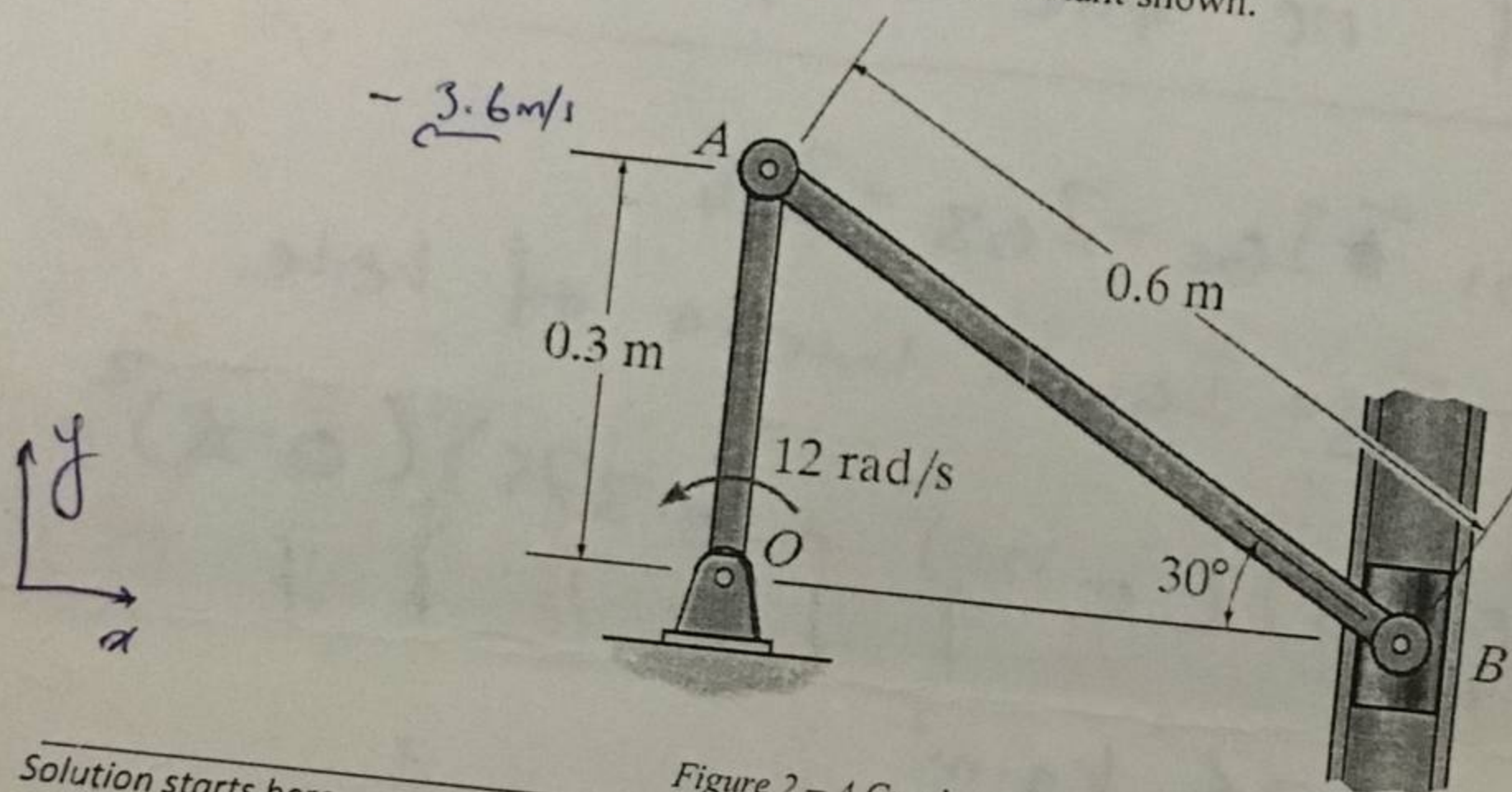
Section: EE

10

Total Marks : 30

Time Allowed: 60 minutes

Q2 If crank OA rotates with an angular velocity of $\omega_0 = 12 \text{ rad/s}$ determine the velocity of piston B and the angular velocity of rod AB at the instant shown. [10]



$$\begin{matrix} i & j & k \\ 0 & 0 & \omega \\ 0.519 & -0.3 & 0 \end{matrix}$$

Solution starts here.....

Figure 2 - A Crank and Piston arrangement

$$\vec{V}_A = \vec{V}_O + \vec{\omega}_0 \times \vec{r}_{A/O}$$

$$\vec{V}_O = 0$$

$$\vec{V}_A = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0.3 \\ 0 \end{bmatrix}$$

$$\vec{V}_A = \begin{bmatrix} -3.6 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}$$

$$\vec{V}_B = \vec{V}_A + \vec{\omega}_{B/A} \times \vec{r}_{B/A}$$

$$\vec{r}_{B/A} = \begin{bmatrix} 0.519 \\ -0.3 \\ 0 \end{bmatrix} \text{ m}$$

$$\begin{bmatrix} 0 \\ V_B \\ 0 \end{bmatrix} = \begin{bmatrix} -3.6 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \omega_{B/A} \end{bmatrix} \times \begin{bmatrix} 0.519 \\ -0.3 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ V_B \\ 0 \end{bmatrix} = \begin{bmatrix} -3.6 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.3\omega_{B/A} \\ 0.519\omega_{B/A} \\ 0 \end{bmatrix}$$

$$0 = -3.6 + 0.3\omega_{B/A}$$

$$\omega_{B/A} = 12 \text{ rad/s} \text{ in the } z \text{ direction}$$

$$V_B = 0.519(12)$$

$$V_B = 6.228 \text{ m/s in the } y \text{ direction}$$



Name: Fawad Arshad

Roll No.: 13L-4411

Section: EE

Total Marks : 30

Time Allowed: 60 minutes

(6)

- Q3 The cart and its load have a total mass of 100 kg. Determine the acceleration of the cart and the normal reactions on the pair of wheels at A and B. Neglect the mass of the wheels. [10]

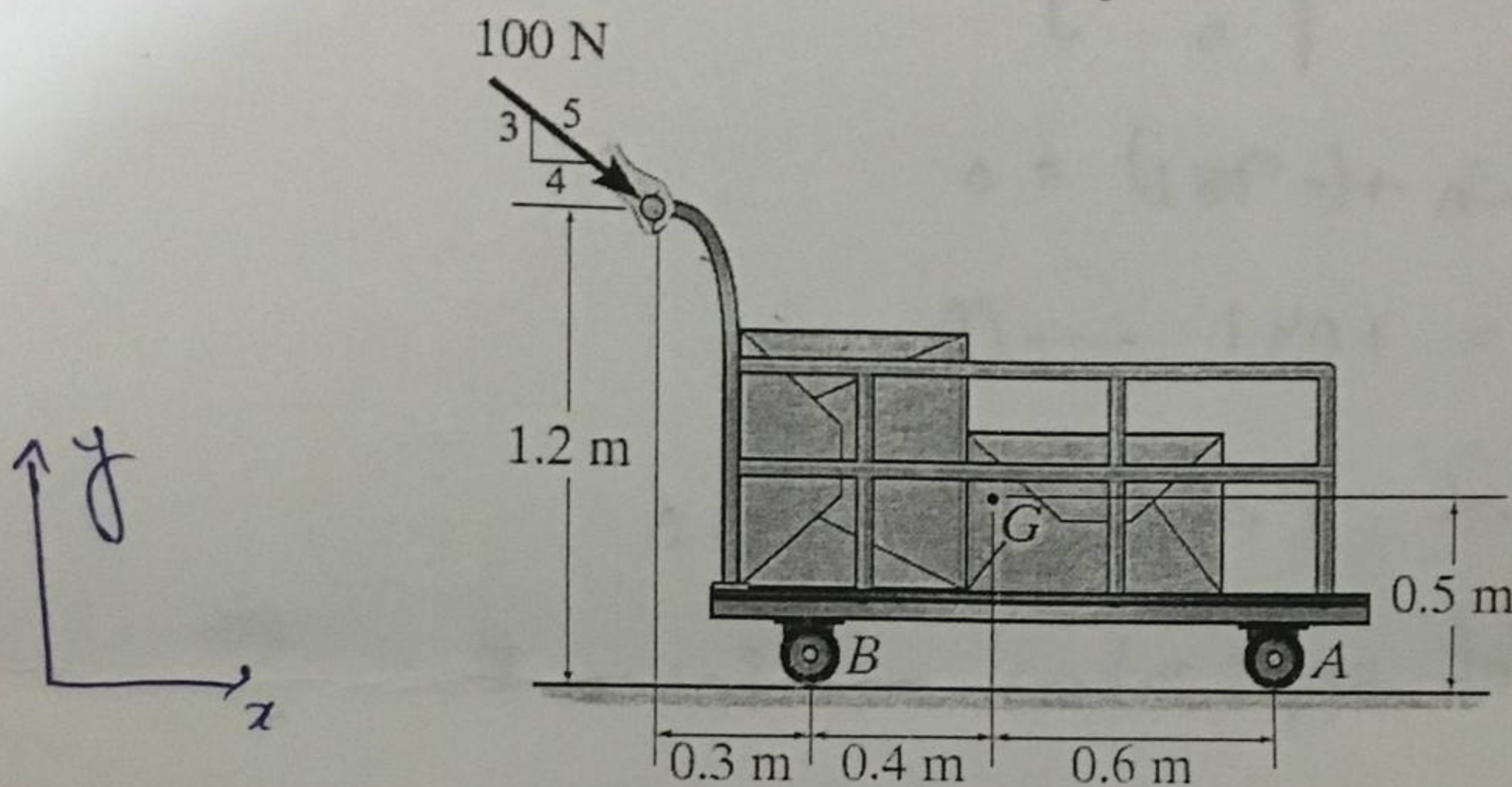
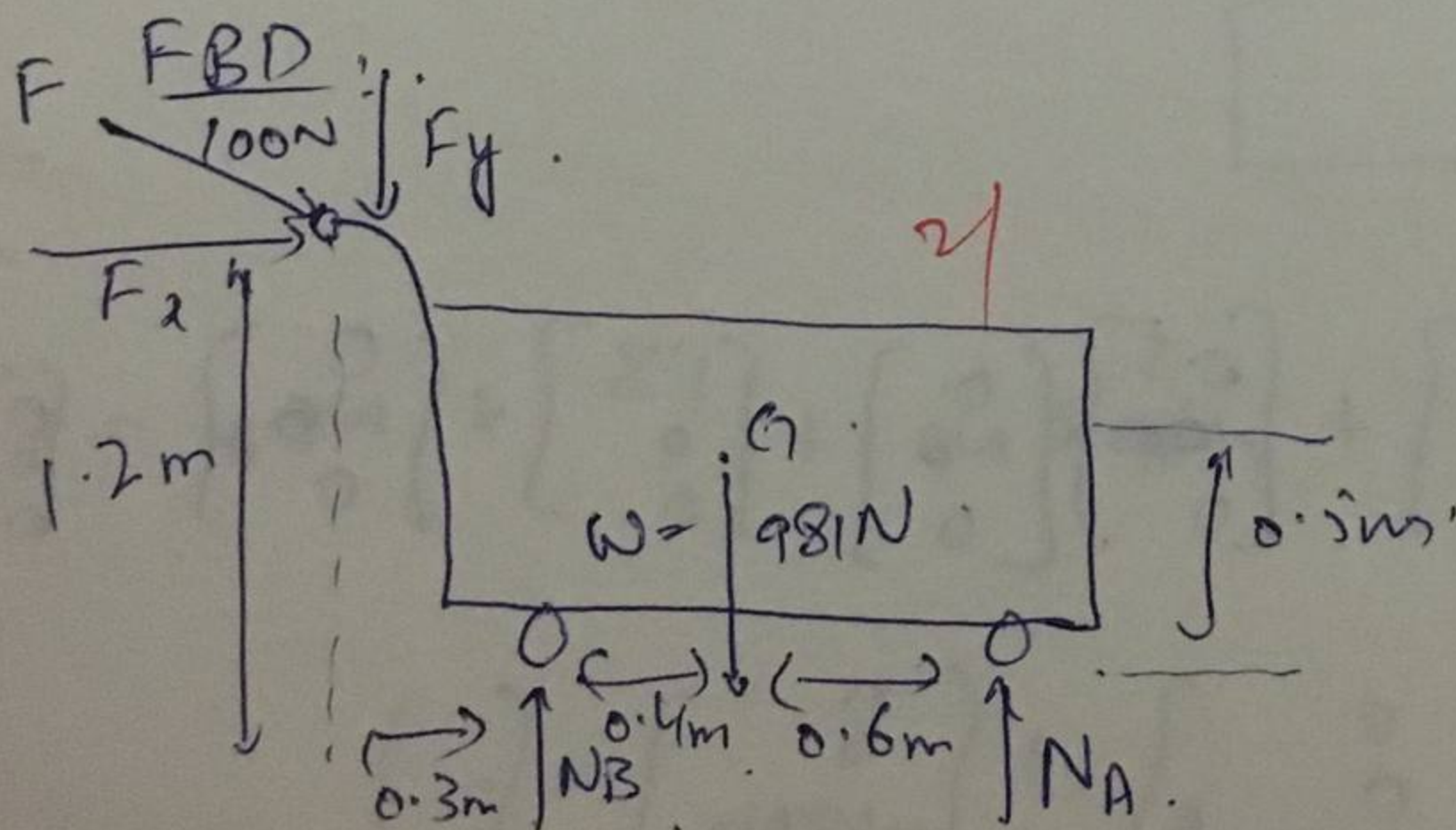


Figure 3 - A Loaded Cart

Solution starts here.....



$$\sum F = ma$$

$$\vec{F} = \begin{bmatrix} 80 \\ -60 \\ 0 \end{bmatrix} \text{ N}$$

$$\vec{N}_B = \begin{bmatrix} 0 \\ N_B \\ 0 \end{bmatrix} \text{ N}$$

$$\vec{N}_A = \begin{bmatrix} 0 \\ N_A \\ 0 \end{bmatrix} \text{ N}$$

$$\begin{bmatrix} 80 \\ -60 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ N_B \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ N_A \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -981 \\ 0 \end{bmatrix} = 100 \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$$

$$W = \begin{bmatrix} 0 \\ -981 \\ 0 \end{bmatrix} \text{ N}$$



Name: Eawad Arshad
Roll No.: 13L-4411

Section: EE

(10)

- Q1 The t-component of the force F is known to be 75 N. Determine the n-component and the magnitude of F . [10]

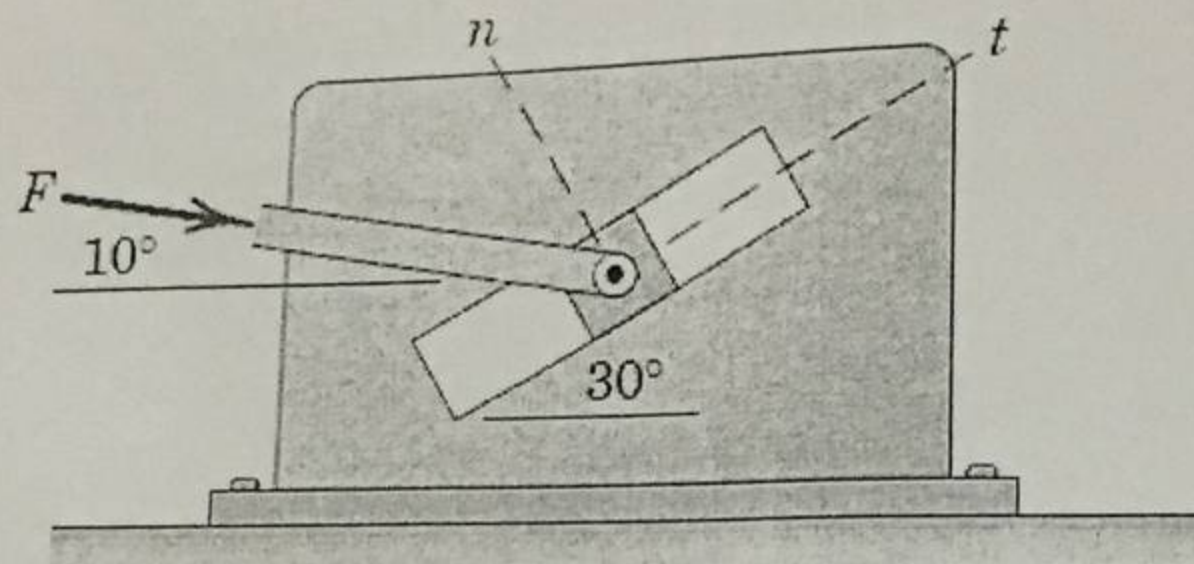
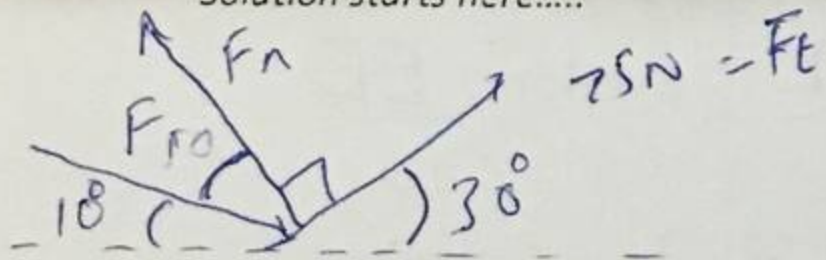
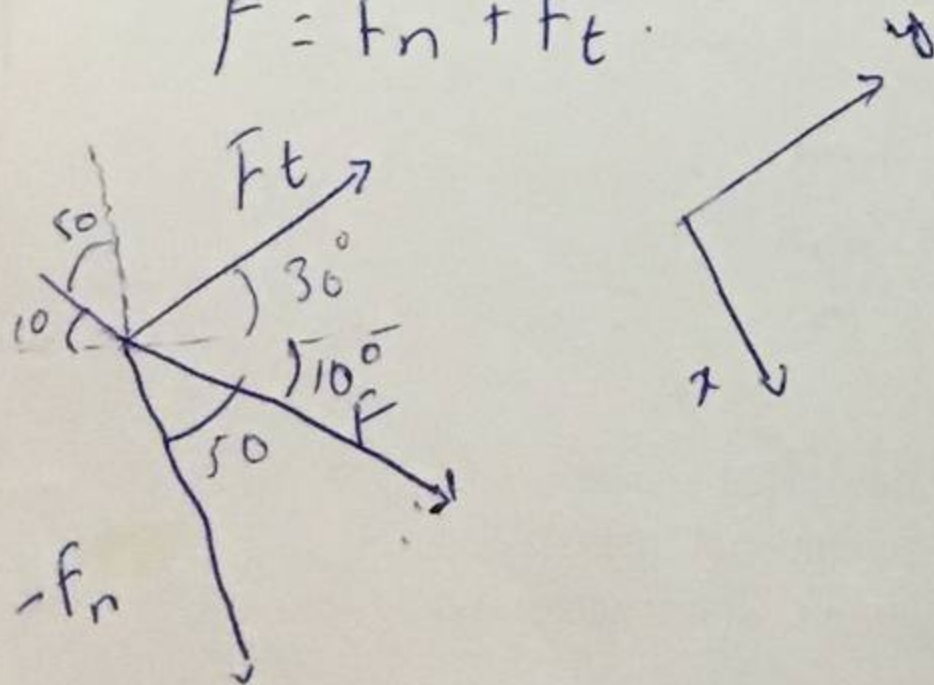


Figure 1 - A Simple Slot Mechanism

Solution starts here.....



$$F = F_n + F_t$$



$$F \sin 40 = 62.932 \text{ N}$$

$$-F_n = 62.932 \text{ N}$$

$$F_n = -62.932 \text{ N}$$

$$F \cos 40 = F_t$$

$$F \cos 40 = 75 \text{ N}$$

$$F = 97.905 \text{ N}$$

FAST-NU
Lahore Campus
Department of Electrical Engineering
Dated: September 14, 2015



ME101 - Basic Mechanical Engineering
Fall 2015
Section EE1
Midterm Examination # 1

Name: Fawad Arshad

Roll No.: 13L-4411

Section: EE

90

Total Marks : 30

Time Allowed: 60 minutes

- Q2 The 120-N force is applied as shown to one end of the curved wrench. If $\alpha = 30^\circ$ calculate the moment of F about the center O of the bolt. Determine the value of α which would maximize the moment about O ; state the value of this maximum moment. [10]

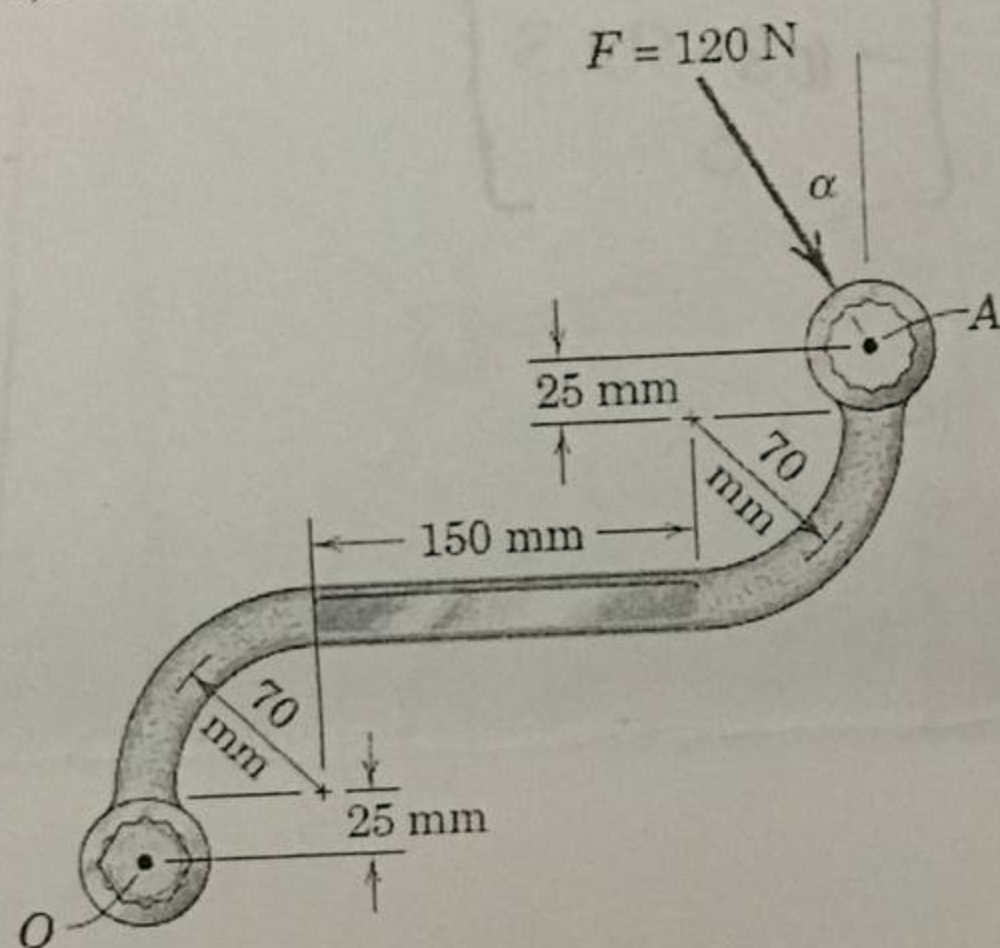
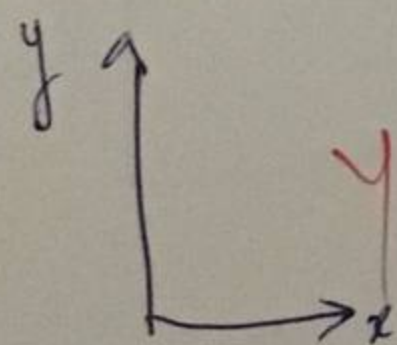


Figure 2 - Curved Wrench

Solution starts here.....

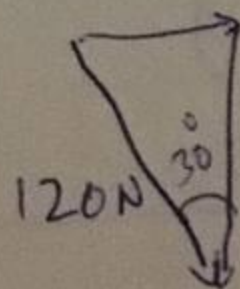


$$\vec{r} = \begin{bmatrix} 70 + 150 + 70 \\ 25 + 70 + 70 + 25 \\ 0 \end{bmatrix} \text{ mm}$$

$$\vec{r} = \begin{bmatrix} 290 \\ 190 \\ 0 \end{bmatrix} \text{ mm}$$

$$\vec{r} = \begin{bmatrix} 0.29 \\ 0.19 \\ 0 \end{bmatrix} \text{ m}$$

$$\vec{F} = \begin{bmatrix} 60 \\ -103.923 \\ 0 \end{bmatrix} \text{ N}$$





Continued Solution of Question 2...

$$\vec{M}_0 = \vec{r} \times \vec{F}$$

$$\vec{M}_0 = \begin{bmatrix} 60 \\ 0.29 \\ 0.19 \end{bmatrix} \times \begin{bmatrix} 60 \\ -103.923 \\ 0 \end{bmatrix}$$

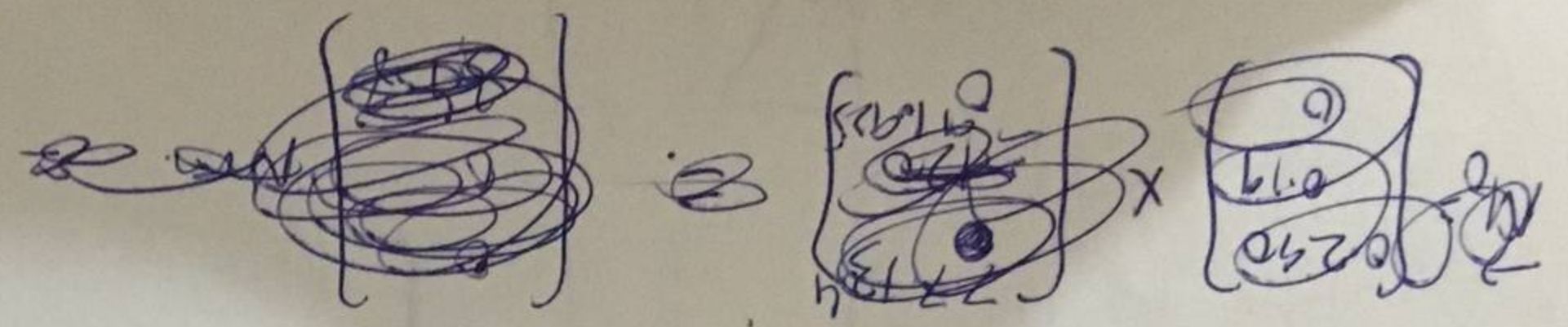
$$\vec{M}_0 = \begin{bmatrix} 60 \\ 0.29 \\ 0.19 \end{bmatrix} \times \begin{bmatrix} 60 \\ -103.923 \\ 0 \end{bmatrix}$$

$$K(-41.537)$$

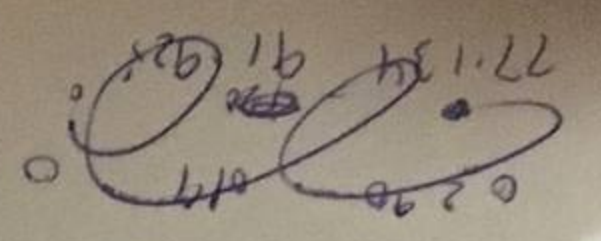
$$\vec{M}_0 = \begin{bmatrix} 0 \\ 0 \\ -41.537 \end{bmatrix} \text{ Nm}$$

direction = -z axis

will maximize the moment as



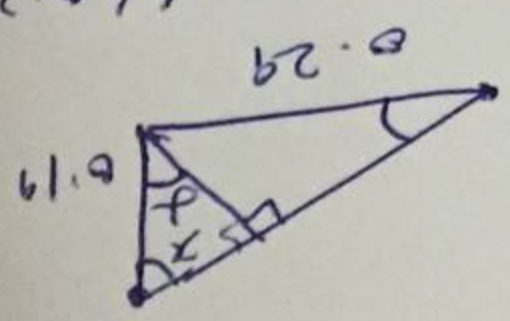
$$\vec{M}_0 \times \vec{F} = (0.29^2 + 0.19^2) \times 120 = 41.604$$



$$\alpha = 90 - 56.768^\circ$$

$$\alpha = 33.2317^\circ$$

$$\alpha = \tan^{-1}\left(\frac{0.29}{0.19}\right)$$



optimum α :



Name: Fawad Ashad

Roll No.: 13L-4411

Section: EE

Total Marks : 30

Time Allowed: 60 minutes

Q3 Three cables are joined at the junction ring C. Determine the tensions in cables AC and BC [10]
caused by the weight of the 30-kg cylinder.

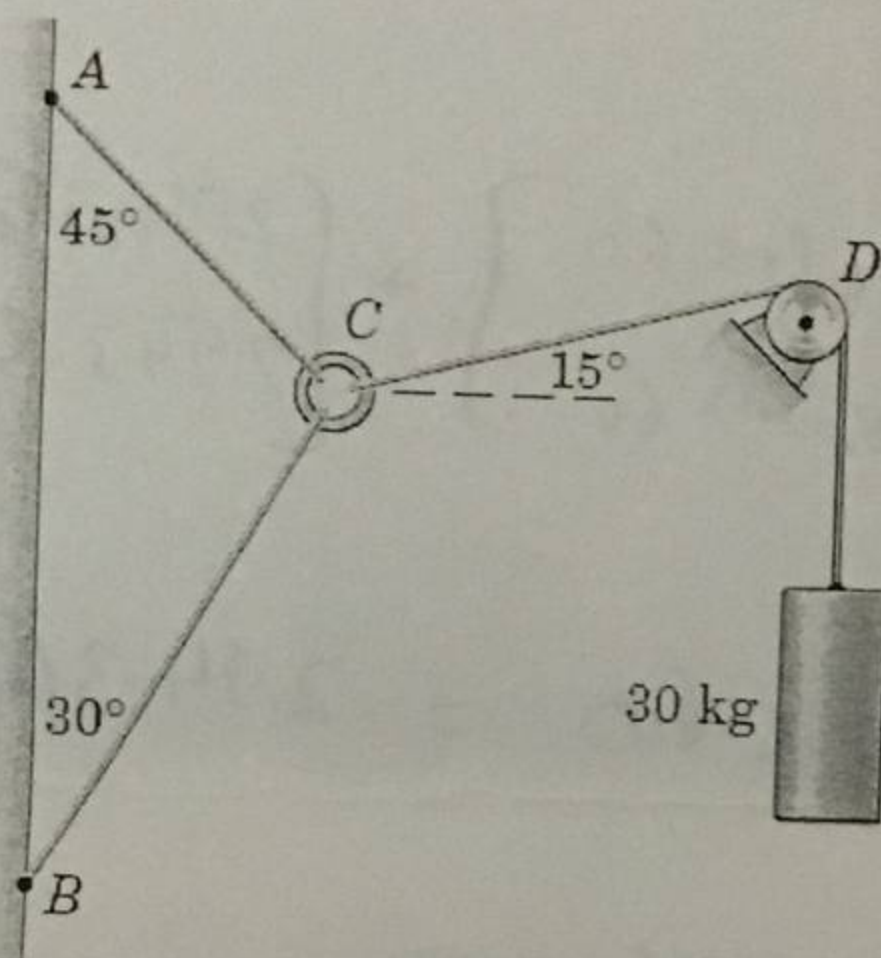
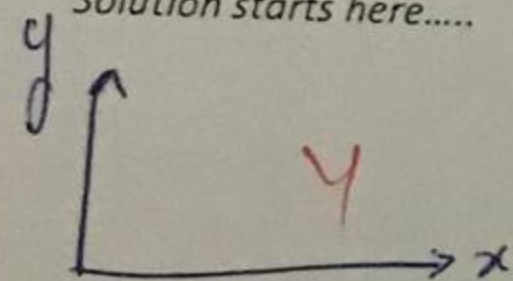
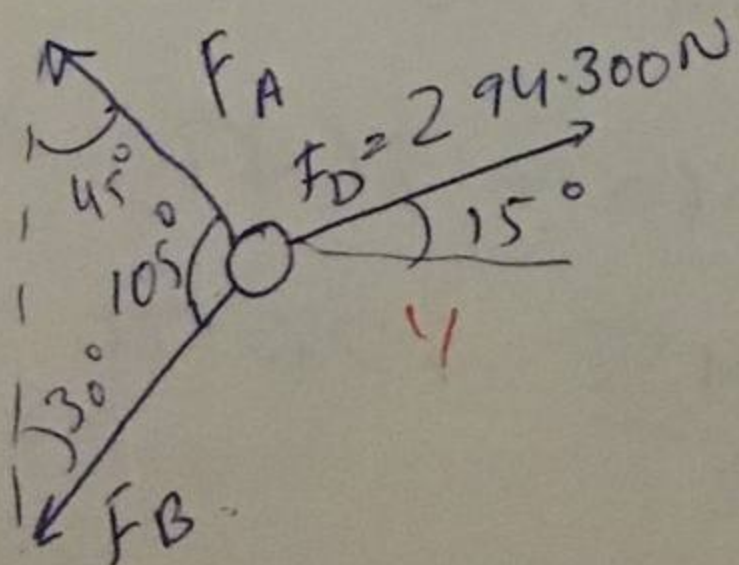


Figure 3 - A Box on an Inclined Plane

Solution starts here.....



let $g = 9.81 \text{ m/s}^2$
 $w = mg$
 $w = 294.300 \text{ N}$



F_A :

$$F_A = \begin{bmatrix} -F_A \cos 45 \\ F_A \sin 45 \\ 0 \end{bmatrix} \text{ N}$$

F_B :

$$F_B = \begin{bmatrix} -F_B \cos 60 \\ -F_B \sin 60 \\ 0 \end{bmatrix} \text{ N}$$

$F_D = \begin{bmatrix} 294.3 \cos 15 \\ 294.3 \sin 15 \\ 0 \end{bmatrix} \text{ N}$



Continued Solution of Question 3...

$$\sum_{all} F = 0$$

$$F_A + F_B + F_D = 0$$

$$\left[\begin{matrix} -F_A \cos 45 \\ F_A \sin 45 \end{matrix} \right] + \left[\begin{matrix} -F_B \cos 60 \\ -F_B \sin 60 \end{matrix} \right] + \left[\begin{matrix} 294.3 \cos 15 \\ 294.3 \sin 15 \end{matrix} \right] = \left[\begin{matrix} 0 \\ 0 \end{matrix} \right]$$

$$F_A \cos 45 + F_B \cos 60 = 294.3 \cos 15 \quad \text{--- (1)}$$

$$\frac{\sqrt{2} F_A}{2} + \frac{F_B}{2} = 294.272$$

$$\sqrt{2} F_A + F_B = 568.544$$

$$F_A \sin 45 - F_B \sin 60 + 294.3 \sin 15 = 0$$

$$F_A \sin 45 - \left[568.544 - \sqrt{2} F_A \right] \sin 60 + 76.170 = 0$$

$$F_A \sin 45 + \frac{\sqrt{2} F_A \sqrt{3}}{2} = \frac{568.544 \sqrt{3}}{2} - 76.170$$

$$F_A \left[\frac{\sqrt{3}}{2} + \frac{\sqrt{2} \sqrt{3}}{2} \right] = 416.203$$

$$F_A = 215.442 \text{ N}$$

$$F_B = 263.862 \text{ N}$$

Name: **Fawad Arshad**
Roll No.: **BEL-4411**
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6

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Q1 The 20-kg sphere rolls down the inclined plane without slipping. Determine the angular acceleration of the sphere and the acceleration of its mass center. [10]

